PAGE 2 — INTRODUCTION TO TISSUES

A tissue is a group of cells that are similar in structure or origin and work together to perform a particular function. In multicellular organisms, specialised cells divide the work between them, making the body more efficient.

Why are tissues important?

- Every cell cannot perform all body functions equally well. Specialised cells are adapted for protection, transport, movement, storage or communication.
- Groups of specialised cells form tissues. Different tissues combine to form organs, and organs work together in organ systems.
- Tissues allow several activities to occur efficiently at the same time. For example, muscle tissue produces movement while nervous tissue coordinates it.
- In plants, tissues provide growth, transport, storage, support and protection.

Levels of organisation

Cell → **tissue** → **organ** → **organ system** → **organism.** Example: muscle cells form muscle tissue; tissues form the heart; the heart works within the circulatory system.

Main classifications

- **Plant tissues:** meristematic and permanent. Permanent tissues include simple, complex and protective tissues.
- **Animal tissues:** epithelial, connective, muscular and nervous tissues.

Key terms

- **Differentiation:** the process by which a cell develops a specialised structure and function.
- **Intercellular space:** space between neighbouring cells.
- **Matrix:** material surrounding cells, especially important in connective tissue.
- **Division of labour:** different cells or tissues perform different tasks.

Plant tissues and animal tissues

Plants require growth at specific regions, mechanical support and transport through xylem and phloem. Animals require movement, rapid coordination, protection and transport throughout the body. Their tissues reflect these different needs.

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PAGE 3 — MERISTEMATIC TISSUES

Meristematic tissue consists of young plant cells that actively divide by mitosis. These cells produce new cells for growth and can later differentiate into permanent tissues.

Characteristics

- Cells are usually small, living and closely packed.
- They have thin cell walls, dense cytoplasm and a prominent nucleus.
- Vacuoles are absent or very small in actively dividing cells.
- Intercellular spaces are generally absent or very limited.
- Repeated cell division supplies new cells to growing regions.

Apical meristem

- Location: tips of roots and shoots.
- Function: increases the length of roots and shoots; this is a type of primary growth.
- Importance: roots grow through soil and shoots grow towards light.

Intercalary meristem

- Location: near nodes or at the bases of leaves, especially in grasses.
- Function: helps elongation of internodes and regrowth after cutting or grazing.
- Example: grass can grow again after being cut because growing tissue remains near its base.

Lateral meristem

- Location: along the sides of stems and roots; examples include vascular cambium and cork cambium.
- Function: increases the thickness or girth of stems and roots; this is secondary growth.
- Importance: helps woody plants develop thicker stems and stronger support.

Exam memory line

Apical = length; Intercalary = growth near nodes or leaf bases; Lateral = thickness.

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PAGE 4 — PERMANENT TISSUES AND PARENCHYMA

Permanent tissues form when meristematic cells mature and differentiate. They acquire a definite shape, size and function and generally lose the ability to divide continuously.

How permanent tissues form

Meristematic cells divide to produce new cells. These cells grow and differentiate, developing structures suited to particular functions.

Classification

- **Simple permanent tissues:** mainly one kind of cell—parenchyma, collenchyma and sclerenchyma.
- **Complex permanent tissues:** different kinds of cells work together—xylem and phloem.
- **Protective tissues:** cover plant surfaces—epidermis and cork.

Parenchyma: structure

- Made of living cells with thin cellulose walls, often rounded, oval or polygonal.
- Intercellular spaces are commonly present.
- Found in soft plant parts such as cortex, pith, fruits and leaves.

Parenchyma: functions and forms

- Stores food and water, fills spaces and can help in healing and repair.
- **Chlorenchyma:** parenchyma containing chloroplasts; it carries out photosynthesis.
- **Aerenchyma:** parenchyma with large air spaces; it helps buoyancy and gas exchange in aquatic plants.

Key point

Parenchyma is a living, usually thin-walled simple permanent tissue. Its function depends on where it is found: storage in some organs, photosynthesis in green parts, and air storage in some aquatic plants.

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PAGE 5 — COLLENCHYMA AND SCLERENCHYMA

Collenchyma and sclerenchyma are simple permanent tissues that provide mechanical support. Their different cell-wall structures make them suitable for flexible or rigid support.

Collenchyma

- Structure: living, elongated cells with unevenly thickened cell walls at the corners, due largely to cellulose and pectin.
- Location: commonly below the epidermis of young stems and in leaf stalks or leaf midribs.
- Function: gives flexible support, allowing young plant parts to bend without breaking easily.
- Example: flexible support in leaf stalks is partly provided by collenchyma.

Sclerenchyma

- Structure: cells usually become dead at maturity and have very thick, lignified walls with a narrow lumen.
- Location: around vascular bundles, in seed coats, nutshells and other hard plant parts.
- Function: provides strength, stiffness and mechanical protection.
- **Fibres** are long and narrow; **sclereids** or stone cells are shorter or irregular and contribute to hardness, such as in pear flesh.

Comparison table

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Feature	Parenchyma	Collenchyma	Sclerenchyma
Cell status	Living	Living	Usually dead at maturity
Wall	Thin	Unevenly thick at corners	Thick and lignified
Main role	Storage, packing, photosynthesis	Flexible support	Strength and rigidity

Memory aid

Parenchyma stores and packs; collenchyma supports flexibly; sclerenchyma strengthens rigidly. In exam answers, describe both structure and function.

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PAGE 6 — COMPLEX AND PROTECTIVE PLANT TISSUES

Complex permanent tissues contain more than one type of cell working together for a common function. Xylem and phloem are the main conducting tissues in plants.

Xylem: water and mineral transport

- Function: carries water and dissolved mineral salts from roots towards stems and leaves; also contributes to support.
- **Tracheids:** elongated cells with thick walls; water passes through pits.
- **Vessels:** tube-like structures made of vessel elements joined end to end; important for water conduction in many flowering plants.
- **Xylem fibres:** provide mechanical strength.
- **Xylem parenchyma:** living cells that store substances and help lateral transport.
- Most xylem elements are dead at maturity; xylem parenchyma is living.

Phloem: food transport

- Function: carries prepared food, especially sugars, from source regions such as mature leaves to sink regions such as roots, fruits and growing tissues.
- **Sieve tubes:** conduct food; their end walls contain sieve plates.
- **Companion cells:** support sieve tube function.
- **Phloem parenchyma:** stores food and helps transport it.
- **Phloem fibres:** provide strength and are dead at maturity.
- Most phloem components are living, with phloem fibres as the important exception.

Xylem versus phloem

- Xylem carries water and minerals; its movement is generally upward from roots to aerial parts.
- Phloem carries food from sources to sinks; movement can occur in different directions in different sieve tubes.

Protective tissues

- **Epidermis:** outer covering of young plant parts; protects against injury and infection and reduces water loss. Many aerial parts have a waxy cuticle.
- **Stomata:** small pores in the epidermis, controlled by guard cells, that allow gas exchange and water vapour loss.
- **Cork:** protective tissue in older stems and roots. Its dead, tightly packed cells have suberin in their walls, reducing water and gas movement.

Importance

Xylem supplies water and minerals needed for photosynthesis and growth. Phloem distributes food from leaves to regions that use or store it.

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PAGE 7 — EPITHELIAL TISSUE

Epithelial tissue covers the body surface, lines internal cavities and organs, and forms many glands. Its cells are closely packed, with very little intercellular space, and rest on a basement membrane.

Functions

- **Protection:** forms barriers against injury, microbes and water loss.
- **Absorption:** takes in useful substances, for example in the small intestine.
- **Secretion:** releases mucus, enzymes, hormones and other substances.
- **Diffusion and filtration:** thin epithelial layers allow exchange in places such as lung alveoli and capillaries.

Types and examples

- **Squamous epithelium:** thin, flat cells; allows diffusion and forms delicate linings, such as alveoli and blood vessels.
- **Stratified squamous epithelium:** multiple layers; protects regions exposed to friction, such as the skin.
- **Cuboidal epithelium:** cube-shaped cells; secretion and absorption, e.g. kidney tubules and gland ducts.
- **Columnar epithelium:** tall cells; absorption and secretion, e.g. parts of the digestive tract.
- **Ciliated epithelium:** cells with hair-like cilia that move material along the surface, such as mucus in parts of the respiratory tract.
- **Glandular epithelium:** specialised for secretion and forms glands.

Special features

- Closely packed cells create a continuous lining or protective barrier.
- The basement membrane supports the epithelium and separates it from underlying tissues.
- Epithelial tissue generally has no blood vessels of its own; nutrients diffuse from nearby connective tissue.

Remember

Thin squamous cells allow diffusion; stratified squamous protects from friction; cuboidal cells help secretion and absorption; columnar cells absorb and secrete; cilia move material; glandular cells produce secretions.

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PAGE 8 — CONNECTIVE TISSUE

Connective tissue supports, binds, protects and connects body parts. Its cells are often spread apart in an extracellular matrix. The matrix may be fluid, flexible or hard according to the tissue's function.

Areolar tissue

- Loose connective tissue found between skin and muscles and around organs, nerves and blood vessels.
- Fills spaces, supports internal organs, binds tissues and helps in repair.

Adipose tissue

- Contains cells specialised for storing fat.
- Found under the skin and around organs such as the kidneys.
- Stores energy, insulates the body and cushions organs against mechanical shocks.

Tendons and ligaments

- **Tendon:** strong, relatively inelastic connective tissue that joins muscle to bone.
- **Ligament:** strong and more elastic connective tissue that joins bone to bone and helps stabilise joints.

Cartilage and bone

- **Cartilage:** firm but flexible tissue found in the nose, outer ear, trachea and at the ends of bones in joints. It provides support and reduces friction.
- **Bone:** hard connective tissue with a matrix strengthened by mineral salts. It supports the body, protects organs, anchors muscles and houses bone marrow.

Blood

- Blood is a fluid connective tissue because its cells are suspended in plasma, a liquid matrix.
- **Red blood cells:** transport oxygen using haemoglobin.
- **White blood cells:** help defend the body against infection.
- **Platelets:** help blood clot.
- **Plasma:** transports nutrients, hormones, wastes and dissolved substances.

Why the matrix matters

The matrix gives connective tissue its properties: a hard matrix makes bone strong, a flexible matrix supports cartilage, and liquid plasma enables blood to transport substances.

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PAGE 9 — MUSCULAR AND NERVOUS TISSUES

Muscular tissue produces movement through contraction. Nervous tissue receives information and transmits signals, helping the body detect changes and coordinate responses.

Skeletal or striated muscle

- Location: attached to bones and used for body movements.
- Structure: long, cylindrical, unbranched fibres with visible stripes called striations; fibres usually contain many nuclei.
- Control: mainly voluntary, meaning it can be consciously controlled.
- Functions: moves the skeleton, maintains posture and helps produce body heat.

Smooth or unstriated muscle

- Location: walls of organs such as the stomach, intestines, bladder and blood vessels.
- Structure: spindle-shaped cells, usually with one central nucleus and no visible striations.
- Control: involuntary.
- Functions: moves food through the digestive tract and changes the diameter of hollow organs and blood vessels.

Cardiac muscle

- Location: wall of the heart.
- Structure: striated, branched cells, usually with a central nucleus, connected by intercalated discs.
- Control: involuntary and rhythmic.
- Function: contracts repeatedly to pump blood through the body.

Muscle comparison

Skeletal: striated and voluntary. Smooth: non-striated and involuntary. Cardiac: striated, branched and involuntary.

Nervous tissue and neuron

- Nervous tissue forms the brain, spinal cord and nerves. Neurons are its main signalling cells; glial cells support and protect them.
- **Cell body:** contains the nucleus and most cell organelles.
- **Dendrites:** receive signals from other cells.
- **Axon:** carries nerve impulses away from the cell body.
- **Myelin sheath:** insulating covering around many axons that helps impulses travel faster.

Reflex action

A reflex is a rapid, automatic response to a stimulus. A typical reflex arc includes a receptor, sensory neuron, a connection in the central nervous system, motor neuron and an effector such as a muscle. Example: withdrawing a hand from a hot object.

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PAGE 10 — FINAL REVISION AND PRACTICE

Use this page as a final revision sheet. Try to answer the questions without looking back at the notes.

High-value differences

- **Meristematic vs permanent:** meristematic cells actively divide; permanent cells are differentiated for specific functions.
- **Xylem vs phloem:** xylem transports water and minerals; phloem transports prepared food.
- **Tendon vs ligament:** tendon joins muscle to bone; ligament joins bone to bone.
- **Skeletal vs smooth muscle:** skeletal muscle is striated and voluntary; smooth muscle is non-striated and involuntary.
- **Neuron vs glial cell:** neurons transmit signals; glial cells support, protect and maintain neurons.

Key terms

- **Differentiation:** development of specialised structure and function.
- **Lignin:** substance that strengthens many plant cell walls.
- **Suberin:** waterproofing substance in cork cell walls.
- **Basement membrane:** supporting layer beneath epithelium.
- **Matrix:** material surrounding cells in connective tissue.
- **Striation:** visible banding in skeletal and cardiac muscle.

Practice questions

- 1. Define tissue and explain the importance of division of labour.
- 2. Name the three types of meristem and state their locations and functions.
- 3. Describe parenchyma, collenchyma and sclerenchyma with one function each.
- 4. Explain the differences between xylem and phloem.
- 5. What are epidermis, stomata and cork? State their roles.
- 6. Name four types of epithelial tissue and give a function or location for each.
- 7. Why is blood called connective tissue? Name its main components.
- 8. Differentiate between tendon and ligament.
- 9. Compare skeletal, smooth and cardiac muscle.
- 10. Draw a labelled neuron and explain dendrites and axon.
- 11. What is a reflex action? Describe a simple reflex arc.

Quick answer check

Apical meristem—length; lateral meristem—thickness; xylem—water/minerals; phloem—food; tendon—muscle to bone; ligament—bone to bone; blood matrix—plasma; neuron—signal transmission.

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